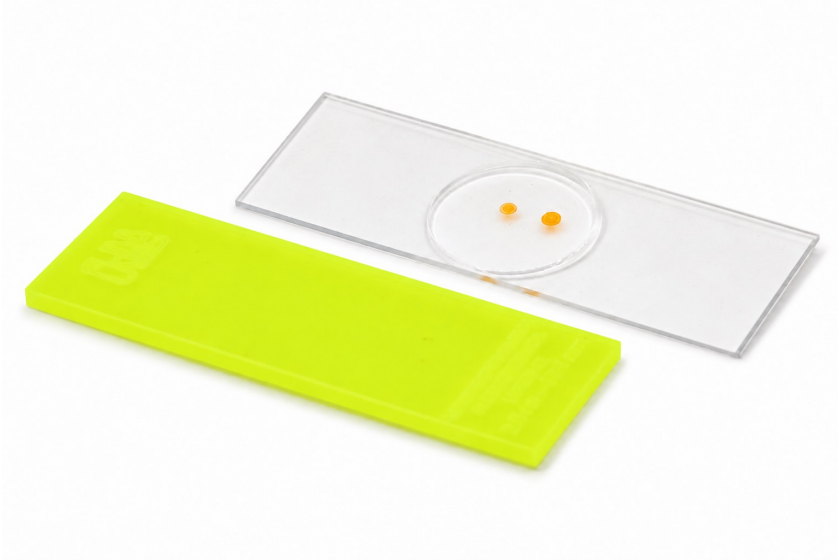
User Manual

# FLIM-Phasors Calibration Kit

**CAL-PH-K-00**



User Manual – Version v1.4.6
26 Oct 2026

# Contents

**1 General Information** **2**
1.1 About This Manual . . . 2
1.1.1 User Qualification . . . 2
1.1.2 Revision History . . . 2
1.2 Symbols and Conventions . . . 2
1.3 Manufacturer Information . . . 2
1.4 Regulatory and Product Classification . . . 3
1.5 Warranty and Liability . . . 3
1.5.1 Warranty . . . 3
1.5.2 Limitation of Liability . . . 3

**2 Product Description** **4**
2.1 Overview . . . 4
2.2 Intended Use . . . 4
2.3 Reasonably Foreseeable Misuse . . . 4
2.4 Technical Specifications . . . 5
2.4.1 Emission Spectra . . . 6
2.4.2 Fluorescence Lifetime Stability . . . 6

**3 Safety Information** **7**
3.1 General Safety Instructions . . . 7
3.2 Handling Precautions . . . 7
3.3 Environmental and Storage Conditions . . . 7

**4 Usage Instructions** **8**
4.1 Scope of Delivery . . . 8
4.2 Unpacking and Inspection . . . 8
4.3 Requirements for Use . . . 8
4.4 Principle of Use . . . 8
4.5 Performing a Calibration with FLIM LABS Software . . . 9
4.6 Troubleshooting . . . 10

**5 Maintenance, Storage, and Disposal** **11**
5.1 Cleaning . . . 11
5.2 Inspection . . . 11
5.3 Storage . . . 11
5.4 Decommissioning and Disposal . . . 11

**A Safety Data Sheet (SDS)** **12**

**B REACH and RoHS Declaration** **14**

# 1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, product classification, accompanying documentation, warranty conditions, and conventions used throughout this document.

## 1.1 About This Manual

This manual provides information for the handling, storage, intended use, and application of the FLIM-Phasors Calibration Kit CAL-PH-K-00.

This document forms an integral part of the product and must remain accessible to the personnel responsible for its use. If the product is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.



Before using the kit, the entire manual must be read carefully and the relevant sections reviewed in detail.

### 1.1.1 User Qualification

The kit is intended for use by qualified personnel familiar with fluorescence microscopy, FLIM measurements, and laboratory sample handling. It is intended exclusively for laboratory and research use. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may use the product only under the supervision of qualified personnel.

### 1.1.2 Revision History

Document Revision: v1.4.6 - 26 Oct 2026

A certified copy of this manual is stored in the manufacturer's technical documentation.

## 1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the accompanying documentation.



Provides important supplementary information.



Indicates procedures that, if not observed, may result in damage to the samples or associated equipment.



Indicates procedures that require particular care in handling, storage, or interpretation of results.

## 1.3 Manufacturer Information


| | |
|---|---|
| Manufacturer: | FLIM LABS S.r.l. |
| Address: | Via della Farnesina, 3 00135 Rome (RM), Italy |
| Telephone: | +39-329 761540 |
| E-mail: | info@flimlabs.com |
| Website: | www.flimlabs.com |




**Figure 1.1:** Product label

## 1.4 Regulatory and Product Classification

The FLIM-Phasors Calibration Kit is a passive laboratory calibration kit. It contains no electronics and does not require electrical power for operation. The product is intended exclusively for research and laboratory use and is not intended for medical, diagnostic, or clinical use. The technical document classifies it as *Research Use Only (RUO)*.

This product does not require CE marking as an electrical or electronic device. However, supporting product documentation may include:

- Safety Data Sheet (SDS)
- RoHS declaration

These documents may be attached in the appendices of this manual.

The product label affixed to the device packaging contains information about product identification and handling instructions. An example product label is shown in Figure 1.1.

## 1.5 Warranty and Liability

### 1.5.1 Warranty

The manufacturer provides warranty coverage in accordance with the applicable commercial terms and conditions. The warranty covers defects in materials and workmanship under normal and intended use.

The warranty is valid only if the product is handled, stored, and used according to this manual. Improper handling, contamination, scratching, unauthorized modification, or use outside the intended purpose voids the warranty.

### 1.5.2 Limitation of Liability

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside laboratory or research applications
- Use for diagnostic, clinical, or medical decision-making
- Improper handling causing contamination or mechanical damage
- Storage outside specified environmental limits
- Misinterpretation of calibration results due to unsuitable measurement conditions

The user is responsible for ensuring that the product is used only within the intended scope and under appropriate laboratory conditions.

# 2 Product Description

## 2.1 Overview

The FLIM-Phasors Calibration Kit is a reference calibration kit for fluorescence lifetime imaging analysis and microscopy (FLIM). It is designed to verify system performance and support calibration prior to experimental measurements.

The kit enables:

- validation of the instrument response function (IRF),
- verification of fluorescence lifetime accuracy and stability,
- qualitative assessment of imaging performance,
- verification of phasor-plot positioning.

The kit contains two reference samples:

1. a fluorescence lifetime reference sample with well-defined monoexponential decay characteristics in the visible spectral range,
2. a biological imaging reference sample for qualitative verification of imaging performance and phasor-space response.

The lifetime reference sample provides a defined fluorescence lifetime of approximately 2.5 ns and is suitable for excitation at 405 nm and 488 nm. The biological reference sample consists of *Convallaria majalis*, commonly used as a qualitative imaging benchmark.

The product is intended exclusively for laboratory and research applications and supports calibration, benchmarking, system validation, and training workflows.

## 2.2 Intended Use

The product is intended for:

- calibration and verification of FLIM and FLIM-Phasors systems,
- determination and validation of the instrument response function (IRF),
- evaluation of fluorescence lifetime accuracy and stability,
- verification of phasor-plot positioning and image quality.

The product is intended exclusively for research use (RUO) and must not be used for medical, diagnostic, or clinical purposes.

## 2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- use for clinical, diagnostic, or medical purposes,
- use of damaged or contaminated reference samples,
- storage or operation outside specified environmental conditions.

## 2.4 Technical Specifications

**Table 2.1:** Technical specifications of the FLIM-Phasors Calibration Kit.

| Item | CAL-PH-K-00 |
|---|---|
| **General** | |
| Product name | FLIM-Phasors Calibration Kit |
| Product code | FLIM-CALB-00 |
| Use classification | Research Use Only (RUO) |
| Intended environment | Indoor laboratory use |
| **Lifetime Reference Sample** | |
| Sample type | Fluorescent lifetime reference sample |
| Dimensions | 75 mm x 25 mm x 3 mm |
| Weight | < 10 g |
| Material | PMMA |
| Fluorescence decay | Monoexponential |
| Reference lifetime | 2.5 ns |
| Excitation wavelengths | 405 nm – 488 nm |
| Emission spectrum at 405 nm | Peak: 495 nm<br>FWHM: 77 nm<br>Spectral centroid: 512 nm |
| **Biological Imaging Reference Sample** | |
| Sample type | Biological imaging reference sample |
| Dimensions | 75 mm x 25 mm x 1 mm |
| Weight | < 10 g |
| Material | Glass |
| Biological specimen | *Convallaria majalis* |
| **Environmental Conditions** | |
| Operating temperature | 0 °C to +40 °C |
| Storage temperature | -20 °C to +70 °C |
| Relative humidity | Up to 80% non-condensing |
| Storage conditions | Clean and dry environment at room temperature |
| Special precaution | Avoid prolonged exposure to intense light when not in use |

### 2.4.1 Emission Spectra

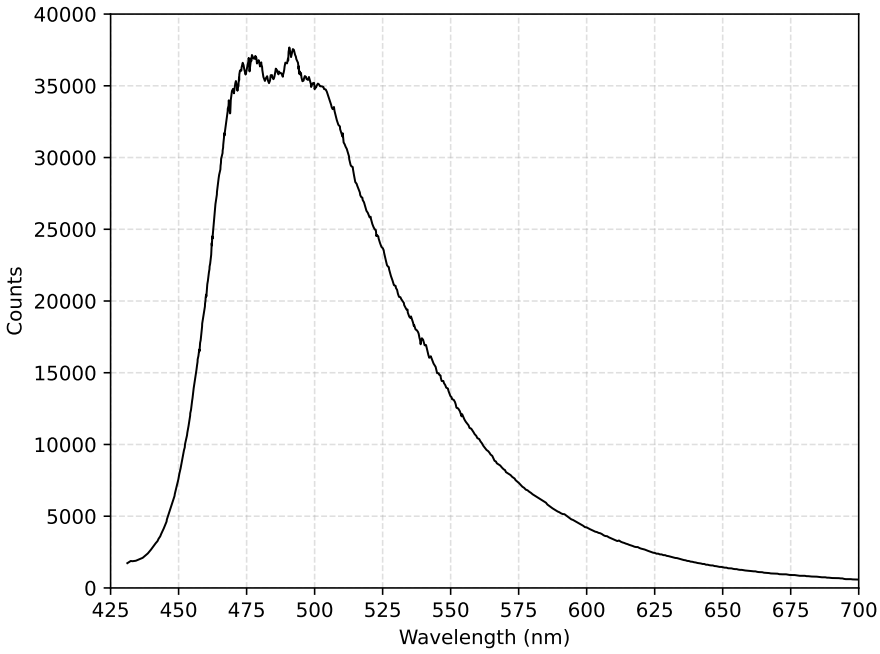



**Figure 2.1:** Emission spectrum of the lifetime reference sample when excited at 405 nm.

### 2.4.2 Fluorescence Lifetime Stability

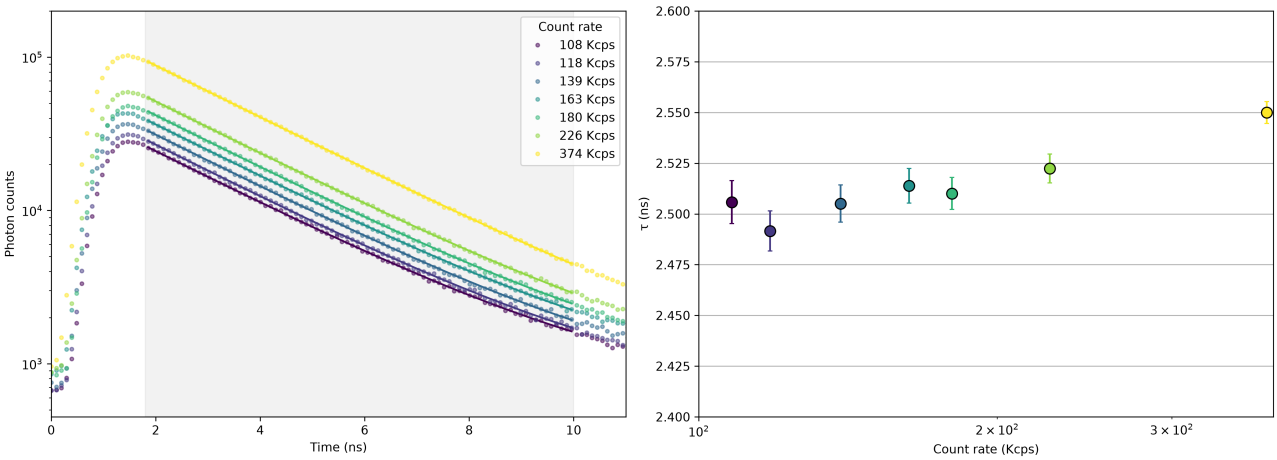



**Figure 2.2:** Measured lifetime $\tau$ of the lifetime reference sample by fitting TCSPC data acquired at different count rates (CPS), tuned via the 488 nm excitation laser power.

# 3 Safety Information

## 3.1 General Safety Instructions

The calibration kit is a passive laboratory product and contains no electrical or electronic hazards. Nevertheless, the reference samples must be handled carefully to avoid damage, contamination, or invalid calibration results.

Use the product only as described in this manual and only in indoor laboratory environments. The installation and storage area must be kept inaccessible to children. These restrictions are stated in the technical document.

## 3.2 Handling Precautions

The calibration kit consists of solid reference samples made of glass and polymer materials. While the product is not classified as hazardous, improper handling may result in mechanical damage or injury.

- Handle all reference slides with care to avoid dropping or impact.
- Do not apply excessive force or bending stress to the samples.
- Inspect samples before use for cracks, chips, or other damage.



Glass components may break if dropped or improperly handled. Broken glass can cause cuts or injury. Handle damaged samples with care and dispose of them according to local regulations.

## 3.3 Environmental and Storage Conditions

Store the kit in a clean and dry environment at room temperature. The technical documentation specifies:

- operating temperature: 0 °C to +40 °C,
- storage temperature: -20 °C to +70 °C,
- relative humidity: up to 80% non-condensing,
- indoor use only,
- do not use near heat sources.

These limits and precautions are taken directly from the technical description.

# 4 Usage Instructions

This chapter describes the kit contents, handling before use, requirements for use, and the basic workflow for using the calibration samples with FLIM LABS software.

## 4.1 Scope of Delivery

The FLIM-Phasors Calibration Kit includes:

- fluorescence lifetime reference sample,
- biological imaging reference sample,
- product documentation.

Verify that all items are present before using the kit.

## 4.2 Unpacking and Inspection

The kit is supplied in protective packaging to prevent damage during storage and transport. Handle the slides carefully during unpacking.
After unpacking:

- inspect the slides for visible damage, cracks, or chips,
- check that the sample surfaces are clean and free from scratches,
- verify that the accompanying documentation is present.

If a slide is damaged, do not use it. Broken glass may cause cuts and must be handled with care.

## 4.3 Requirements for Use

Before using the kit, ensure that:

- the microscope or FLIM system is correctly installed and operational,
- the required excitation wavelength is available,
- the detector and acquisition settings are suitable for the selected sample,
- the environmental conditions comply with the limits specified in Section 3.3,
- the software used for acquisition and analysis is installed and configured.

The reference samples should be used only in indoor laboratory environments.

## 4.4 Principle of Use

The calibration kit provides reference samples for checking and calibrating fluorescence lifetime measurements.
The lifetime reference sample has a known mono-exponential fluorescence decay and can be used to verify lifetime accuracy or generate calibration data. The biological imaging reference sample provides structured fluorescence features for qualitative assessment of image quality and phasor-space behavior.
During use, the reference slide is placed in the microscope sample holder and measured under defined acquisition conditions. The measured data can then be compared against the expected sample behavior or used by the software to generate calibration information.



For reliable calibration, use the same imaging and illumination conditions for the reference measurement and the subsequent sample measurement whenever possible.

## 4.5 Performing a Calibration with FLIM LABS Software

### Placing the Reference Slide

Place the selected reference slide into the microscope sample holder. Ensure that the slide is seated securely and that the measurement area is clean and in focus.

Avoid touching the optical surface of the slide. If the slide is contaminated or scratched, the measurement result may be affected.

### Use with Spectroscopy Py

Use the lifetime reference sample to acquire reference data in Spectroscopy Py.

General workflow:

1. Place the lifetime reference slide in the optical setup.
2. Switch to *Spectroscopy Mode* and select *Phasors Ref* as the calibration type.
3. Select the appropriate acquisition settings.
4. Enter the known decay constant in the field Tau (ns) .
5. Acquire the reference data.
6. Save or export the generated calibration file.

For detailed instructions, refer to the Spectroscopy Py software manual on the website.

### Use with FLIM Imager

Use the lifetime reference sample in the Calibration mode of FLIM Imager.

General workflow:

1. Place the lifetime reference slide in the microscope.
2. Focus the sample and adjust the illumination intensity.
3. Select *Calibration* tab in the FLIM analysis modes.
4. Enter the known decay constant in the field Tau (ns) .
5. Start the calibration acquisition.
6. Apply the calibration data in Phasors mode.

For detailed instructions, refer to the FLIM Imager software manual on the website.



If the detected signal is above the detector saturation threshold, reduce the excitation intensity using neutral density filters or by de-focusing the beam.

## 4.6 Troubleshooting

**Table 4.1:** Common issues when using the calibration kit.

| Issue | Possible Cause | Recommended Action |
|---|---|---|
| Unexpected lifetime value | Incorrect software settings or calibration parameters | Verify the acquisition settings, synchronization settings, and entered reference lifetime. |
| Weak fluorescence signal | Insufficient excitation intensity or incorrect focus | Check the focus position and excitation path. Increase illumination only within safe detector limits. |
| Detector saturation | Excitation intensity too high | Reduce excitation intensity using ND filters or defocus the beam. |
| Poor image quality | Incorrect focus, dirty slide, or unsuitable imaging settings | Refocus the sample, inspect the slide surface, and verify microscope settings. |
| Inconsistent calibration results | Different settings between reference and sample measurement | Use matching illumination, detector, acquisition, and reconstruction settings for calibration and sample measurement. |
| Visible scratches, cracks, or chips | Mechanical damage to the slide | Do not use the damaged slide. Handle broken glass carefully and dispose of it according to local regulations. |

# 5 Maintenance, Storage, and Disposal

## 5.1 Cleaning

Clean the reference samples only when necessary and using methods suitable for optical or microscopy slides.

- Remove loose dust carefully, e.g. using compressed air.
- Do not use abrasive materials.
- Do not use aggressive solvents unless confirmed compatible with the sample material.

## 5.2 Inspection

Periodically inspect the kit components to ensure proper condition and usability.

- Check sample surfaces for scratches, contamination, or other damage.
- Verify that packaging remains intact and provides adequate protection.
- Ensure that accompanying documentation is complete and legible.

## 5.3 Storage

Store the kit under appropriate environmental conditions to preserve sample integrity.

- Keep in a clean and dry environment.
- Store at room temperature unless otherwise specified.
- Ensure compliance with the specified storage limits.
- Avoid prolonged exposure to intense light when not in use.

## 5.4 Decommissioning and Disposal

If the kit is no longer suitable for use due to damage, contamination, or end of life, remove it from service to prevent unintended reuse.

Dispose of the components in accordance with applicable local regulations for laboratory consumables and material waste. Disposal procedures may differ depending on the material (e.g., PMMA or glass).

# A Safety Data Sheet (SDS)

This appendix summarizes the Safety Data Sheet (SDS) for the FLIM-Phasors Calibration Kit CAL-PH-K-00. The product is a passive laboratory calibration kit intended for research use only.

## 1. Product Identification

| | |
|---|---|
| Product Name: | FLIM-Phasors Calibration Kit |
| Product Code: | FLIM-CALB-00 |
| Use: | Calibration kit for fluorescence microscopy (RUO) |
| Manufacturer: | FLIM LABS S.r.l. |
| Address: | Via della Farnesina, 3, 00135 Rome (RM), Italy |
| Email: | info@flimlabs.com |
| Emergency Phone: | +39-329 761540 |

## 2. Hazard Identification

- Not classified as hazardous under REACH/CLP.
- No flammable, toxic, or reactive components.
- Minimal risk under normal laboratory handling conditions.

## 3. Composition / Information on Ingredients

**Table A.1:** Composition / Information on Ingredients

| Component | CAS No. | Concentration | Hazard Classification |
|---|---|---|---|
| PMMA (Polymethyl Methacrylate) | 9011-14-7 | 75–100% | Not hazardous |
| Borosilicate Glass | 65997-17-3 | 0–25% | Not hazardous |
| *Convallaria majalis* (plant tissue) | N/A | <1% | Not hazardous |

## 4. First-Aid Measures

- **Ingestion:** Rinse mouth with water. Seek medical advice if discomfort occurs.
- **Eye contact:** Rinse with water if particles enter the eye.
- **Skin contact:** Wash with soap and water if contaminated.

## 5. Fire-Fighting Measures

- Suitable extinguishing media: water, foam, dry powder, $CO_2$.
- No hazardous combustion products under normal conditions.

## 6. Accidental Release Measures

- Handle carefully to avoid breakage.
- Dispose of broken slides according to local glass disposal regulations.

## 7. Handling and Storage

- Handle with care to avoid scratches or breakage.
- Store in a clean, dry environment at room temperature.

- Avoid prolonged exposure to intense light.

## 8. Exposure Controls / Personal Protection

- Use standard laboratory precautions.
- Protective gloves recommended when handling broken slides.
- Safety glasses recommended if there is a risk of breakage.

## 9–16. Additional Information

- Physical state: solid, non-volatile, non-flammable.
- Stability: chemically stable under normal conditions.
- Toxicological and ecological effects: no known hazards.
- Disposal: follow local regulations for laboratory materials.
- Transport: non-hazardous.
- Regulatory status: Research Use Only (RUO), no CE marking required.

**Document Information:**

| | |
|---|---|
| Prepared by: | Felix Sturm |
| Reviewed by: | Alessandro Rossetta |
| Approved by: | Alessandro Rossetta |
| Date: | 15/02/2026 |
| Revision: | 0 (Initial release) |

# B REACH and RoHS Declaration

## REACH & RoHS Declaration

### FLIM-Phasors Calibration Kit

Product Name: FLIM-Phasors Calibration Kit
Manufacturer: Flimlabs S.r.l.
via della Farnesina, 3
00135 Rome (RM), Italy
Date: 15/02/2026

## REACH Compliance

- The product does not contain substances of very high concern (SVHC) listed in the latest European Chemicals Agency (ECHA) candidate list above the threshold of 0.1% weight/weight.
- Components: PMMA slides, glass slide, plant tissue (Convallaria majalis). All materials comply with REACH requirements.

## RoHS Compliance

- The product does not contain restricted hazardous substances listed in EU Directive 2011/65/EU (RoHS) above the maximum concentration values.
- The product contains no electronic components, lead, mercury, cadmium, hexavalent chromium, PBB, or PBDE.

## Conclusion

This product is in full compliance with REACH and RoHS regulations applicable to its material composition.

Authorized Signatory:



[Alessandro Rosetta, CEO]

**Figure B.1:** Reach and RoHS Declaration